Customer Behavior & Sales Analysis Report

1. Introduction

This project aims to analyze customer shopping behavior and sales performance to extract actionable business insights. By combining data preprocessing, database management, and interactive visualization, the project provides a complete pipeline from raw data to decision-making support.

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2. Dataset Description

The dataset was loaded from a CSV file containing customer shopping behavior data. It includes:

- Customer demographics (Age, Gender)
- Transaction details (Category, Purchase Amount)
- Behavioral attributes (Purchase Frequency, Subscription Status)
- Customer feedback (Review Rating)
- Marketing indicators (Discount Applied)

The dataset required preprocessing to ensure data quality and consistency before analysis.

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3. Methodology

3.1 Data Preprocessing (Python ETL)

Using Pandas, a structured ETL pipeline was implemented:

- **Data Inspection**
  - Previewed data using head()
  - Checked structure using info()
  - Generated statistical summaries using describe()
- **Handling Missing Values**
  - Identified null values using isnull().sum()

- Missing values in **Review Rating** were filled using **median per category**, ensuring more accurate imputation

Data Cleaning

- Standardized column names:
  - Converted to lowercase
  - Replaced spaces with underscores
- Renamed:
  - purchase_amount_(usd) → purchase_amount

🔹 **Feature Engineering**

- Created **age_group** using quantile-based segmentation (pd.qcut)
  - Young Adult
  - Adult
  - Middle-aged
  - Senior
- Converted purchase frequency into numeric format:
  - Weekly → 7 days
  - Monthly → 30 days
  - Quarterly → 90 days
- Removed redundant column:
  - promo_code_used (duplicate of discount_applied logic)

3.2 Database Integration

The cleaned dataset was stored in a MySQL database using:

- mysql-connector-python
- SQLAlchemy

Steps included:

- Establishing connection to MySQL
- Creating database: customer_behavior
- Writing cleaned data into table: customer

This step ensures scalability and efficient querying for further analysis.

3.3 SQL Analysis

SQL was used to:

- Aggregate revenue and sales by category
- Segment customers by age group and subscription status
- Prepare structured outputs for dashboard visualization

3.4 Data Visualization (Dashboard)

An interactive dashboard was built using Power BI to present insights visually.

- **KPIs**
  - Total Customers: **3.9K**
  - Average Purchase Amount: **$59.76**
  - Average Review Rating: **3.75**
- **Visual Components**
  - Revenue by Category
  - Sales by Category
  - Revenue by Age Group
  - Sales by Age Group
  - Subscription Distribution
- **Filters (Slicers)**
  - Gender
  - Category
  - Shipping Type

- Subscription Status

- **4. SQL Analysis & Business Insights**

After storing the cleaned dataset in the customer_behavior database, advanced SQL queries were used to extract meaningful business insights.

🔹 4.1 Revenue by Gender

```
select gender, sum(purchase_amount) as revenue

from customer

group by gender;
```

📌 Insight:

This analysis compares spending behavior between male and female customers, helping businesses:

Identify the dominant revenue segment

Tailor marketing strategies accordingly

🔹 4.2 High-Value Discount Customers

```
select customer_id, purchase_amount

from customer

where discount_applied = 'Yes'

and purchase_amount >= (select avg(purchase_amount) from customer);
```

📌 Insight:

Some customers use discounts but still spend above average

➡️ These are high-value customers, not just bargain seekers

➡️ Ideal for targeted promotions and loyalty programs

🔹 4.3 Top Rated Products

```
select item_purchased, round(avg(review_rating),2) as avg_rating
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

📌 Insight:

Identifies the top 5 products by customer satisfaction

➡️ These products can be:

Promoted more

Used as flagship items

🔹 4.4 Shipping Type Impact

```
select shipping_type, round(avg(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type;
```

📌 Insight:

Compares spending between shipping preferences

➡️ Helps determine if faster shipping leads to higher spending

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🔹 4.5 Subscription vs Spending

```
SELECT subscription_status,
    round(AVG(purchase_amount), 2) AS avg_spend,
    COUNT(customer_id) AS total_customers,
```

```
    ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status;
```

📌 Insight:

Evaluates whether subscribed customers are more valuable

➡️ Key for:

Subscription model decisions

Retention strategies

🔹 4.6 Discount Sensitivity by Product

```
select item_purchased,
round(sum(case when discount_applied = 'yes' then 1 else 0 end)/count(*) * 100,2) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

📌 Insight:

Shows which products are most dependent on discounts

➡️ Useful for pricing and promotion strategies

🔹 4.7 Customer Segmentation

```
case
when previous_purchases = 1 then 'new'
when previous_purchases between 2 and 10 then 'returning'
else 'loyal'
```

end

📌 Insight:

Customers are segmented into:

New

Returning

Loyal

➡️ Helps in:

Personalized marketing

Retention campaigns

🔹 4.8 Top Products per Category

row_number() over(partition by category order by count(customer_id) desc)

📌 Insight:

Identifies top 3 products in each category

➡️ Useful for:

Inventory planning

Product placement

🔹 4.9 Repeat Buyers & Subscription Behavior

where previous_purchases > 5

📌 Insight:

Checks if loyal customers are more likely to subscribe

➡️ Helps evaluate subscription effectiveness

🔹 4.10 Revenue by Age Group

```
select sum(purchase_amount), age_group
from customer
group by age_group;
```

📌 Insight:

Shows which age groups generate the most revenue

➡️ Supports targeted marketing strategies

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5. Business Recommendations

📈 **Increase Subscription Conversion**

- Offer loyalty programs or exclusive discounts
- Use targeted email campaigns

🛍️ **Focus on High-Performing Categories**

- Expand **Clothing** inventory
- Increase marketing investment in top categories

📉 **Improve Underperforming Categories**

- Re-evaluate pricing and product quality in:
  - Footwear
  - Outerwear

🎯 **Target High-Value Customers**

- Focus marketing campaigns on **Young Adults**
- Personalize offers based on purchase behavior

⭐ **Improve Customer Experience**

- Address factors affecting review ratings
- Optimize shipping options and service quality

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6. Conclusion

This project demonstrates a complete data pipeline, starting from raw data preprocessing using Python, storing structured data in SQL databases, and delivering business insights through an interactive Power BI dashboard.

The analysis highlights key opportunities in subscription growth, category optimization, and customer targeting, enabling data-driven decision-making.